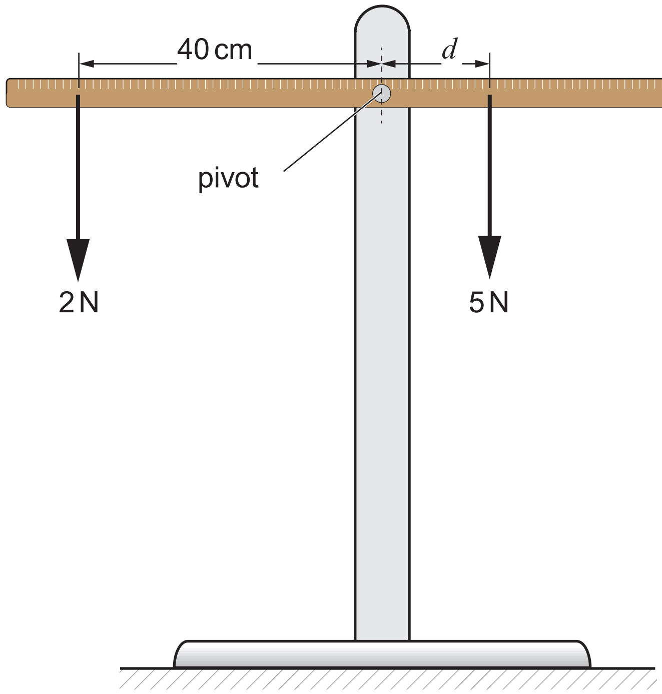
Examiner only

4. Some students are investigating the principle of moments. They use a metre ruler and some different weights. The ruler is pivoted at its centre. To begin with they set up their experiment as shown below. They vary the position of the 5 N weight until the ruler is balanced.



(a) (i) State the principle of moments. [1]

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....

(ii) Use an equation from page 2 to calculate the moment of the 2 N weight about the pivot. Give your answer in N cm. [2]

Moment = ... N cm

Examiner only

(iii) Use the equation:

$$\text{distance} = \frac{\text{moment}}{\text{force}}$$

to calculate the distance that the 5 N weight must be from the pivot in order to balance the ruler. [2]

Distance = ... cm

(b) The students replace the 2 N weight with a 4 N weight. Jade suggests that to balance the ruler they will have to halve the distance that the 5 N weight is from the pivot. Explain whether you agree with her suggestion. [2]

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